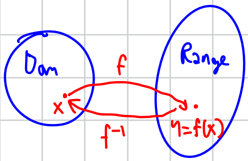
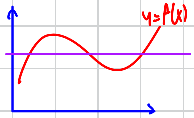
7.6 Inverse Trigonometric Functions

Recall 7.1 : f is 1-1 if $x_1, x_2 \in Dom(f)$, $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$

→ Graph of f satisfies horizontal line test





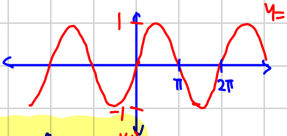
Not 1-1

$f^{-1}(y) = x$ if and only if $y = f(x)$

- $f^{-1}(f(x)) = x \quad \forall x \in Dom(f) = Range(f^{-1})$

  $f(f^{-1}(y)) = y = \forall y \in Dom(f^{-1}) = Range(f)$

Consider $y = \sin x$ on natural domain $(-\infty, \infty)$

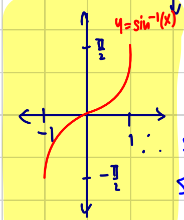


Horizontal line test → Not 1-1

But if restrict to be $[-\frac{\pi}{2}, \frac{\pi}{2}]$ → 1-1 $\therefore f^{-1}$ exists

Range $(f) = [-1, 1] = Dom(f^{-1})$

Notation : $f^{-1}(x) = \sin^{-1}(x) = \arcsin(x)$



$y = \sin^{-1}(x)$, $x \in [-1, 1]$ if and only if $\sin y = x$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

Given $x \in [-1, 1]$, $\sin^{-1}(x)$ is an angle $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, whose sine is equal to x

$\therefore \sin(\sin^{-1}(x)) = x \quad \forall x \in [-1, 1]$

$\sin^{-1}(\sin x) = x = \forall x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

Properties of $\sin^{-1}(x)$

1. Continuous on $[-1, 1]$
2. $\sin^{-1}(-x) = -\sin^{-1}(x) \quad \forall x \in [-1, 1]$
3. $(\sin^{-1} x)' = ?$

$\therefore y = \sin^{-1} x \rightarrow \sin y = x$

$\cos y \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}}$

$\cos y = \sqrt{1 - \sin^2 y}$ since $y \in [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow \cos y \geq 0$

$\therefore \int \frac{1}{\sqrt{1 - x^2}} dx = \sin^{-1}(x) + C$

$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \int \frac{1}{\sqrt{a^2 - a^2u^2}} (a\, du) = \int \frac{1}{\cancel{a}} \frac{1}{\sqrt{1-u^2}} \cancel{a}\, du = \int \frac{1}{\sqrt{1-u^2}} du = \sin^{-1}(u) + C = \sin^{-1}\left(\frac{x}{a}\right) + C$

Problems

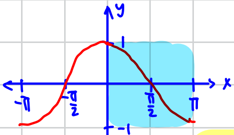
7.6/3 a) $\sin^{-1}(-\frac{1}{2}) = y \rightarrow y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$; $\sin y = -\frac{1}{2} \quad \therefore y = -\frac{\pi}{6}$

$\sin^{-1}(-\frac{1}{2}) = -\sin^{-1}(\frac{1}{2})$

b) $\sin^{-1}(\frac{1}{\sqrt{2}}) = \frac{\pi}{4}$

7.6/13 $\lim_{x \to 1^-} \sin^{-1}(x) = \sin^{-1}(1) = \frac{\pi}{2}$

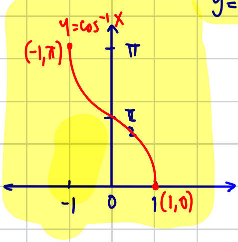
Inverse of cos X   Natural domain : $(-\infty,\infty)$



Not satisfy horizontal line test

$\rightarrow$ Restricted domain : $[0,\pi]$, Range = $[-1,1]$

$\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad$ $\parallel$ Range$(\cos^{-1})$ $\quad$ $\parallel$ Dom$(\cos^{-1})$



$y=\cos^{-1}(x)$, $x\in[-1,1]$ if and only if $y\in[0,\pi]$, $\cos y = x$

Properties of $\cos^{-1}(x)$

1. It is continuous on $[-1,1]$
2. $\cos^{-1}(-x)=\pi-\cos^{-1}(x)$ $\quad \forall x\in[-1,1]$

   $[\cos(\pi-\cos^{-1}x)=-\cos(\cos^{-1}x)=-x]$

   $\therefore \cos^{-1}(-x)=\pi-\cos^{-1}x$
3. $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$ $\quad \forall x\in[-1,1]$

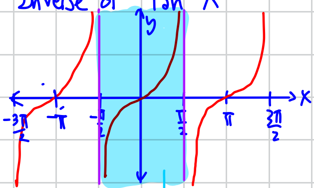
   $y=\cos^{-1}(x)\rightarrow y\in[0,\pi]$, $\cos y = x$

   $\frac{\pi}{2}-y\in[-\frac{\pi}{2},\frac{\pi}{2}]$, $\sin(\frac{\pi}{2}-y)=\overset{1}{\cancel{\sin\frac{\pi}{2}}}\cos y-\overset{0}{\cancel{\cos\frac{\pi}{2}}}\sin y=\cos y$

   $\therefore \sin^{-1}(x)=\sin^{-1}(\cos y)=\sin^{-1}(\sin(\frac{\pi}{2}-y))=\frac{\pi}{2}-y=\frac{\pi}{2}-\cos^{-1}x$

   $\therefore \sin^{-1}(x)+\cos^{-1}(x)=\frac{\pi}{2}$
4. $\frac{d}{dx}(\cos^{-1}x)=\frac{d}{dx}(\frac{\pi}{2}-\arcsin x)=-\frac{1}{\sqrt{1-x^2}}$, $x\in[-1,1]$

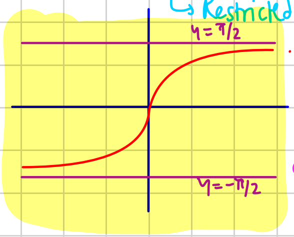
Inverse of tan X



Natural domain: $(-\infty,\infty)\setminus\{\pm\frac{\pi}{2},\pm\frac{3\pi}{2},\ldots\}$

Range(tan) $=(-\infty,\infty)$ = Dom$(\tan^{-1})$

$\hookrightarrow$ Restricted area: $(-\frac{\pi}{2},\frac{\pi}{2})$ = Range $(\tan^{-1})$



Vertical Asymptote $\rightarrow$ Horizontal Asymptote

Properties of $\tan^{-1}x$

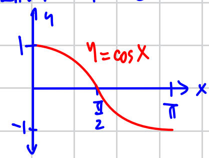
1. Contin on $(-\infty,\infty)$
2. $\lim_{x\to\infty}\tan^{-1}(x)=\frac{\pi}{2}$, $\lim_{x\to-\infty}\tan^{-1}(x)=\frac{-\pi}{2}$
3. $\frac{d}{dx}(\tan^{-1}x)$: $y=\tan^{-1}x$; $x\in(-\infty,\infty)$, $y\in(-\frac{\pi}{2},\frac{\pi}{2})\rightarrow \tan y = x$

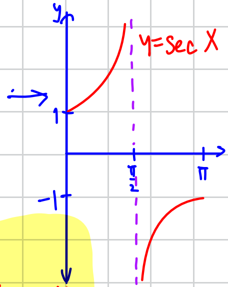
   $(\arctan x)'=\frac{dy}{dx}=\frac{1}{\frac{dx}{dy}}=\frac{1}{\sec^2 y}=\frac{1}{1+\tan^2 y}=\frac{1}{1+x^2}$

   $\therefore (\tan^{-1}x)'=\frac{1}{1+x^2}$; $x\in(-\infty,\infty)$

   $\int\frac{1}{1+x^2}dx=\tan^{-1}(x)+C$

Inverse of $\sec x$

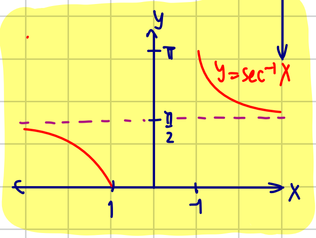




$y=\sec x \longrightarrow 1-1$ with restricted domain of $\sec x$: $[0,\pi]-\{\frac{\pi}{2}\}$ = range of $\sec^{-1}$

$\text{Range}(\sec x) = (-\infty,-1]\cup[1,\infty) = \text{dom}(\sec^{-1})$

$y=\sec^{-1}x,\ x\in(-\infty,-1]\cup[1,\infty) \longleftrightarrow y\in[0,\pi]\setminus\{\frac{\pi}{2}\},\ \sec y = x$



Properties of $\sec^{-1}x$

$(0,\frac{\pi}{2})$: $\sec x>0$, $\tan x>0$
$(\frac{\pi}{2},\pi)$: $\sec x<0$, $\tan x<0$

1. continuous on its domain
2. $(\sec^{-1}x)' = ?$ $\quad \frac{dy}{dx} = 1\Big/\frac{dx}{dy}$

$x=\sec y$ $\qquad = \frac{1}{\sec y\tan y} = \frac{1}{|x|\sqrt{x^2-1}}$

$\therefore \int \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1}(x)+C$

Check $(\sec^{-1}|x|)' = \frac{1}{||x||\sqrt{|x|^2-1}}\cdot |x|' \quad \frac{x}{|x|} \quad = \frac{1}{x\sqrt{x^2-1}}$ $\quad \left(\frac{x}{|x|^2}=\frac{1}{x}\right)$

$\therefore \int\frac{dx}{x\sqrt{x^2-1}} = \sec^{-1}|x|+C$

Quiz next wk

7.6/79 $\int\frac{dx}{(x+1)\sqrt{x^2+2x+1-1}} = \int\frac{dx}{(x+1)\sqrt{(x+1)^2-1}} = \int\frac{du}{u\sqrt{u^2-1}} = \sec^{-1}|x+1|+C$

Chapter 8 : Tricks to do $\int f(x)\,dx$

8.2 Integration by parts

Scenario: $\int f(x)\,\boxed{g(x)\,dx}$ → easily integrated ex. $g(x)=e^x \to \int g(x)dx = e^x+C$

$f(x)$ → ugly but breakable

$f(x) = x^{929}+423x^{2025}+1113$
$\to f'(x) = 929x^{928}+423\cdot 2025\,x^{2024}$

Recall $(f(x)g(x))' = f(x)g'(x)+f'(x)g(x)$

$\int (f(x)g(x))'\,dx = \int f(x)g'(x)\,dx + \int f'(x)g(x)\,dx$

$\qquad \overset{\shortparallel}{f(x)g(x)}$ $\qquad\qquad \int g'(x)\,dx = g(x)+C$

$\therefore \int f(x)g'(x)\,dx = f(x)g(x) - \underline{\int f'(x)g(x)\,dx}$

original ; easily integrated ; new one nicer

More user-friendly notation

Let $u=f(x)$, $v=g(x)$
$du=f'(x)dx$, $dv=g'(x)dx$ $\Big\}$ $\int u\,dv = uv-\int v\,du$

e.g. $\int xe^x\,dx$ $\qquad$ Let $u=x$, $v=e^x$
$=\int u\,dv$ $\qquad du = 1\,dx$, $dv=e^x\,dx$
$= xe^x-\int e^x dx = xe^x - e^x + C$

e.g. 2 $\int x \ln x \, dx$: Let $u = \ln x \rightarrow du = \frac{1}{x} dx$

$\underbrace{x\,dx}_{dv}$ $\quad v = \frac{x^2}{2} \rightarrow dv = x\,dx$

$= \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$

$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$

e.g. 3 $\int x \sin x \, dx$: $\quad u = x \quad du = 1\,dx$

$v = -\cos x \rightarrow dv = \sin x \, dx$

$= -x\cos x - \int -\cos x \, dx$

$= -x\cos x + \sin x + C$

e.g. 4 $\int \cos x \, e^x \, dx$ $\quad$ Let $u = \cos x \rightarrow du = -\sin x \, dx$

$= e^x \cos x - \int -e^x \sin x \, dx$ $\quad v = e^x \rightarrow dv = e^x dx$

$= e^x \cos x + \int e^x \sin x \, dx$

$\int e^x \sin x \, dx = \sin x \, e^x - \int e^x \cos x \, dx$ $\quad u = \sin x \rightarrow du = \cos x \, dx$

$= e^x \sin x - \int e^x \cos x \, dx$ $\quad v = e^x \rightarrow dv = e^x dx$

$\therefore \int \cos x \, e^x \, dx = e^x (\sin x + \cos x) - \int \cos x \, e^x \, dx$

$\therefore \int \cos x \, e^x \, dx = \frac{e^x}{2} (\sin x + \cos x) + C$

Ugliness ranking
① $\ln x \rightarrow$ Ugly breakable / crackable
② Polynomials,
③ e^x a^x, $\sin x$, $\cos x$

8.3 Trig $\int f(x)\,dx$

$\int (\sin x)^m (\cos x)^n \, dx$; $m, n \geq 0$ integers

Case 1 m is odd ; $m = 2k+1$

$u = \cos x$

$\int \sin^m x \cdot \cos^n x \, dx = \int (\sin x)^{2k} \cdot \cos^n x \cdot \sin x \, dx = \int (1-\cos^2 x)^k \cdot \cos^n x \cdot -d(\cos x) = \int -(1-u^2)^k \cdot u^n \, du$

Case 2 n is odd ; $n = 2l+1$

$u = \sin x$

$\int \sin^m x \cdot \cos^n x \, dx = \int \sin^m x \cdot (\cos x)^{2l} \cdot \cos x \, dx = \int \sin^m x \cdot (1-\sin^2 x)^l \cdot d(\sin x) = \int u^m \cdot (1-u^2)^l \, du$

Case 3 both m, n are even $\quad (m = 2k, \ n = 2l)$

$\int \sin^m x \cos^n x \, dx = \int (\sin^2 x)^k \cdot (\cos^2 x)^l \, dx = \int \left(\frac{1-\cos 2x}{2}\right)^k \left(\frac{1+\cos 2x}{2}\right)^l dx$

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha$

$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \sin\beta\cos\alpha$

$\therefore \sin\alpha\cos\beta = \frac{\sin(\alpha+\beta) + \sin(\alpha-\beta)}{2}$

$\cos\alpha\cos\beta = \frac{\cos(\alpha+\beta) + \cos(\alpha-\beta)}{2}$

$\sin\alpha\sin\beta = \frac{\cos(\alpha-\beta) - \cos(\alpha+\beta)}{2}$

Product of sine and cosine

$m, n \geq 0$

$\int \sin mx \cos nx \, dx = \frac{1}{2} \int \sin(mx+nx) + \sin(mx-nx) \, dx$

$= \frac{1}{2} \left[-\frac{\cos(mx+nx)}{m+n} - \frac{\cos(mx-nx)}{m-n} \right] + C \quad (m \neq n)$

$\int \sin mx \sin nx \, dx = \frac{1}{2} \int \cos(mx-nx) - \cos(mx+nx) \, dx$

$= \frac{1}{2} \left[\frac{\sin(mx-nx)}{m-n} - \frac{\sin(mx+nx)}{m+n} \right] + C$

Form $\int \tan^m x \sec^n x \, dx$

$m=1, n=0$: $\int \tan x \, dx = \ln|\sec x| + C$

$m=0, n=1$: $\int \sec x \, dx = \ln|\sec x + \tan x| + C$

Case 1: n even $\rightarrow n = 2k$; $k \geq 1$

$u = \tan x$

$$\int \tan^m x \sec^n x \, dx = \int \tan^m x \sec^{2k} x \, dx = \int \tan^m x (\sec^{2k-2} x) \sec^2 x \, dx = \int \tan^m x (\tan^2 x + 1)^{k-1} \cdot d(\tan x) = \int u^m (u^2+1)^{k-1} du$$

Case 2: $n = 0, m \geq 2$

$m=2$: $\int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx = \tan x - x + C$

$m \geq 3$: $\int \tan^m x \, dx = \int \tan^{m-2} x \cdot \tan^2 x \, dx = \int \tan^{m-2} x \cdot (\sec^2 x - 1) \, dx = \int \tan^{m-2} x \underbrace{(\sec^2 x \, dx)}_{d(\tan x)} - \int \tan^{m-2} dx$

$= \int u^{m-2} du - \int \tan^{m-2} x \, dx$

$= \frac{\tan^{m-1} x}{m-1} - \int \tan^{m-2} x \, dx$ ทำไปเรื่อยๆ จน $m = 1$ หรือ 2

Case 3: $m = 0, n \geq 2$

$n = 2$: $\int \sec^2 x \, dx = \tan x + C$

$n \geq 3$, odd: $n = 2k+1$

$\heartsuit = \int \sec^{2k+1} dx = \int \sec^{2k-1} x \cdot \underbrace{\sec^2 x \, dx}_{d(\tan x)}$

$= \sec^{2k-1} x \cdot \tan x - \int \tan x \, d(\sec^{2k-1} x)$

$= \sec^{2k-1} x \cdot \tan x - \int \tan x (2k-1) \sec^{2k-2} x (\sec x \tan x) \, dx$

$= \sec^{2k-1} x \cdot \tan x - (2k-1) \int \tan^2 x \sec^{2k-1} x \, dx$

$= \sec^{2k-1} x \tan x - (2k-1) \int \sec^{2k-1} x \cdot (\sec^2 x - 1) \, dx$

$2k \cdot \heartsuit$ $= \sec^{2k-1} x \tan x + (2k-1) \int \sec^{2k-1} x \, dx$

$\heartsuit$ $= \left(\sec^{2k-1} x \tan x + (2k-1) \int \sec^{2k-1} x \, dx\right) / 2k$ $\rightarrow$ ทำต่อไปเรื่อยๆ

Case 5 $n = 1$

FEEL

NOT MEMORIZE

$\underbrace{\int \tan^m x \sec x \, dx}_{\star} = \int \tan^{m-1} x \cdot \underbrace{\sec x \tan x \, dx}_{d(\sec x)}$

$= \tan^{m-1} x \sec x - \int \sec x (m-1) \tan^{m-2} x (\sec^2 x) \, dx$

$= \tan^{m-1} x \cdot \sec x - (m-1) \int \tan^{m-2} x (\tan^2 x + 1) \sec x \, dx$ $\nearrow$ $\underbrace{\int \tan^m x \sec x \, dx}_{\star} + \int \tan^{m-2} x \sec x \, dx$

$m \int \tan^m x \sec x \, dx = \tan^{m-1} x \cdot \sec x - (m-1) \int \tan^{m-2} x \sec x \, dx$

8.5 Integral of rational functions $\int \frac{P(x)}{Q(x)} dx$, $P(x) \leq Q(x)$, polynomials

$$\int \frac{1}{x^2 - x - 12} dx = \int \frac{1}{7} \left(\frac{1}{x-4} - \frac{1}{x+3} \right) dx = \frac{1}{7} \left(\int \frac{1}{x-4} dx - \int \frac{1}{x+3} dx \right) = \frac{1}{7} \ln \left| \frac{x-4}{x+3} \right| + C$$

$\int \boxed{\frac{99x + 423}{x^2 - x - 12}} dx = \frac{A^{117}}{x-4} + \frac{B^{-18}}{x+3}$ for all $x \neq 4, -3$ $\rightarrow A \ln|x-4| + B \ln|x+3| + C$

Homework: 7.6, 8.2, 8.3

$99x + 423 = A(x+3) + B(x-4)$

$x = 4 \rightarrow 7A = 99 \cdot 4 + 423$, $x = -3 \rightarrow -7B = 99 \cdot (-3) + 423$